

PARUL UNIVERSITY - FACULTY OF ENGINEERING& TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

6th SEM B.TECH. PROGRAMME

Cryptography (203105381)

ACADEMIC YEAR 2020-21

Chapter:3 BLOCK CIPHERS AND THE DATAENCRYPTION STANDARD

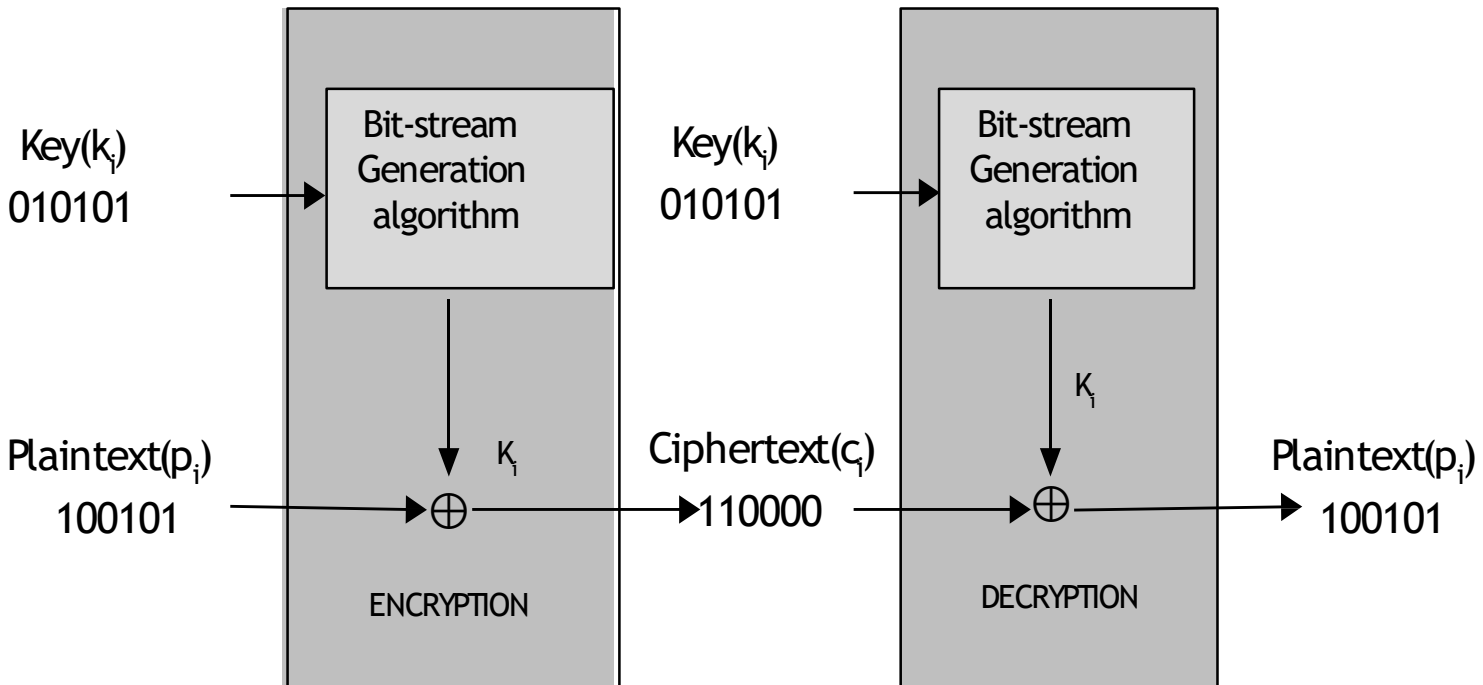
Syllabus:

Block Ciphers and the Data Encryption Standard: Block Cipher Principles, Data Encryption Standard (DES), Deferential and Linear Cryptanalysis, Block Cipher Design Principles, Block Cipher Operation, RC4.

Stream Cipher::

A stream cipher is one that encrypts a digital information stream one bit or one byte data at a time.
Examples:
Vigenère cipher, Ceasar Cipher, Vernam cipher.

BLOCK CIPHERS::





Diffusion and Confusion:

- Claude Elwood Shannon introduced the concepts diffusion and uncertainty.
- Diffusion" =
  In order to avoid attempts to deduce the secret, the diffusion process aims to ma ke the statistical relationship between the plaintext and the cipher text as comple x as possible.
- Confusion" = Confusion seeks to make the relationship as complex as possible between cipher text and the value of the encryption key."

Thus, even though the attacker can get some idea of the cipher text statistics, then the manner in which the key was used to generate cipher text is so complicated that it is difficult to locate the key. This is done with the aid of a complex algorithm for substitution.

DES- Standard of Data Encryption:

A symmetric-key block cipher published by the National Institute of Standards and Technology is the Data Encryption Standard (DES) (NIST).

• NIST issued a request for proposals for a national symmetric-key cryptosystem in 1973.

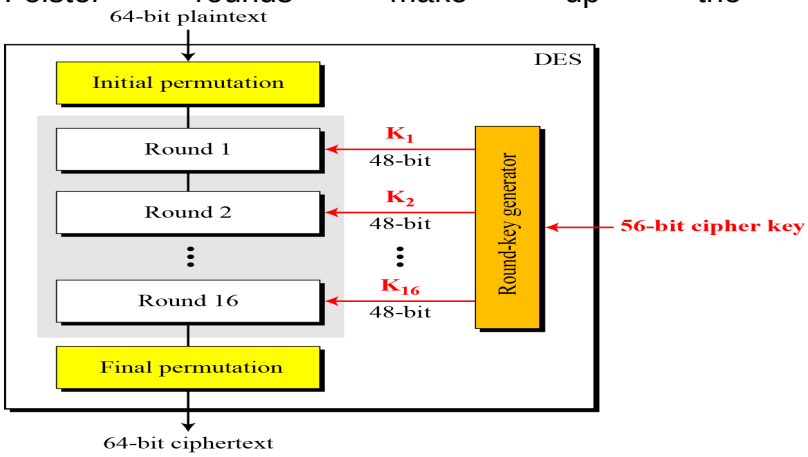
A proposal from IBM was approved as DES, a modification of a project named Lucifer.

In March 1975, DES was published in the Federal Register as a draft of the Federal Standard for Processing Information (FIPS).

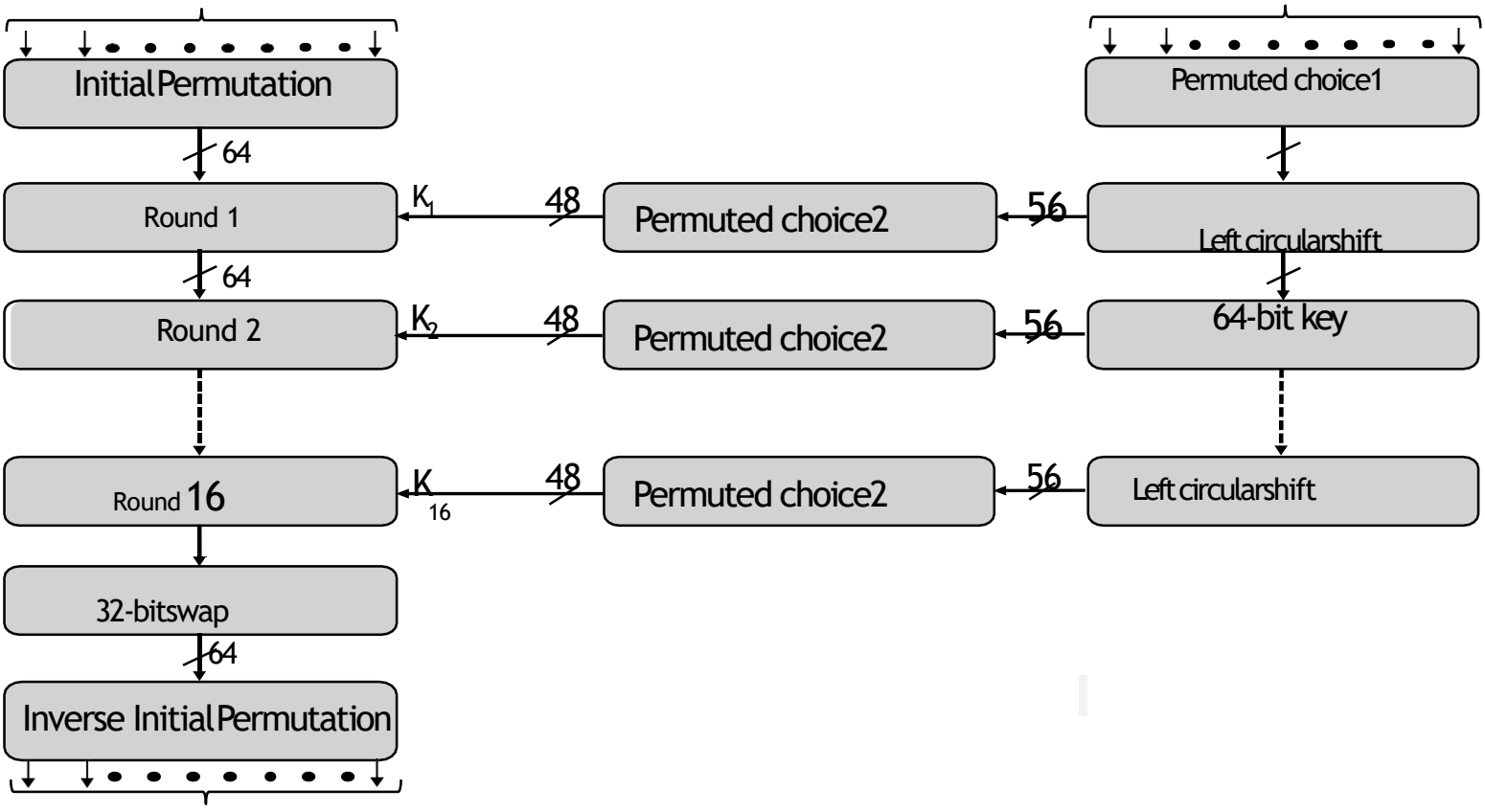
DES STRUCTURE:



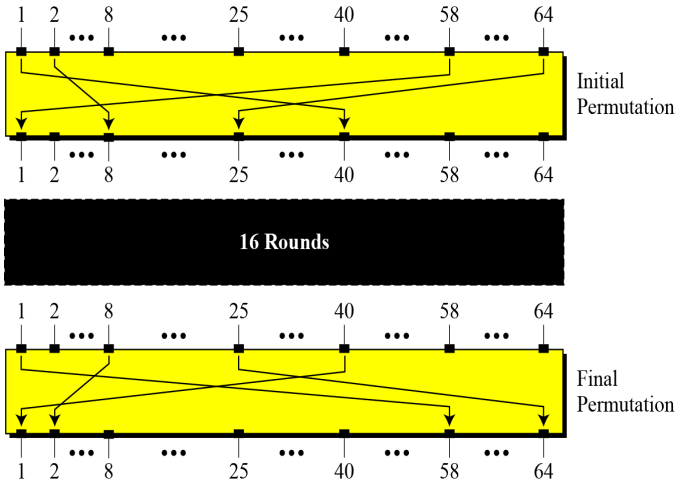
Two permutations (P-boxes), which we call initial and final permutations, and sixteen Feistel rounds make up the encryption process.



DES Encryption Algorithm:



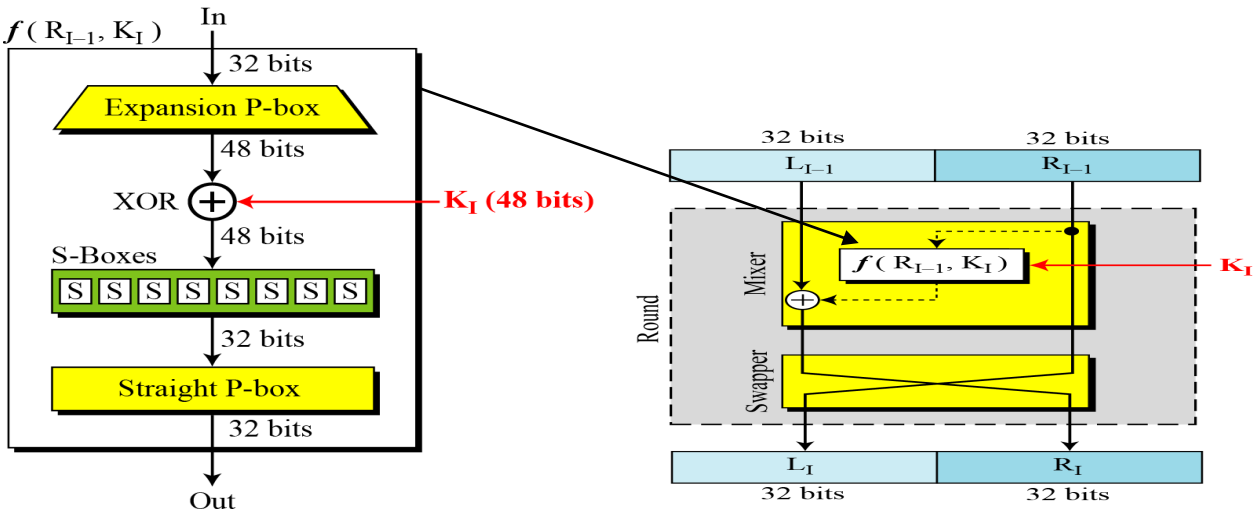
Initial and Final Permutations in DES:



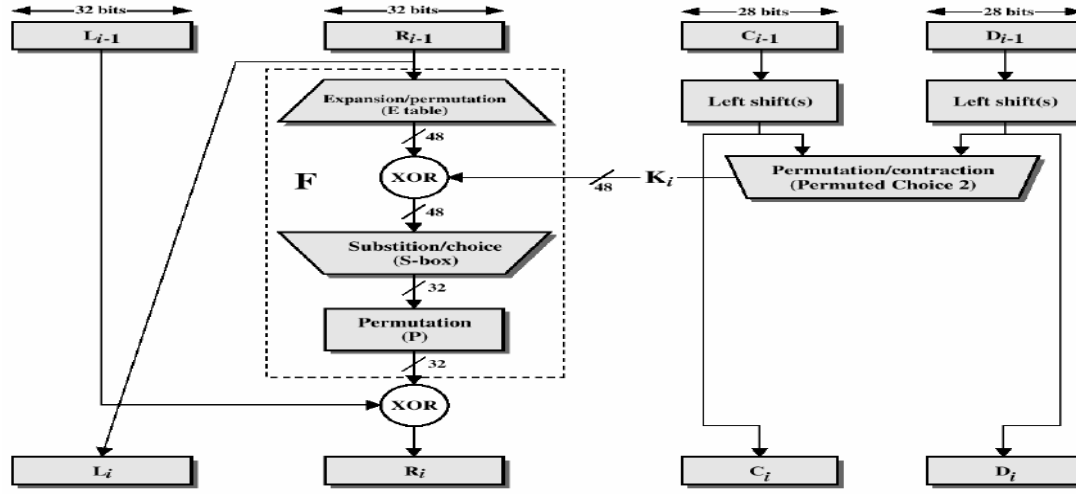
Initial Permutation								Final Permutation							
58	50	42	34	26	18	10	02	40	08	48	16	56	24	64	32
60	52	44	36	28	20	12	04	39	07	47	15	55	23	63	31
62	54	46	38	30	22	14	06	38	06	46	14	54	22	62	30
64	56	48	40	32	24	16	08	37	05	45	13	53	21	61	29
57	49	41	33	25	17	09	01	36	04	44	12	52	20	60	28
59	51	43	35	27	19	11	03	35	03	43	11	51	19	59	27
61	53	45	37	29	21	13	05	34	02	42	10	50	18	58	26
63	55	47	39	31	23	15	07	33	01	41	09	49	17	57	25

DES Function:

The DES feature is the heart of DES. In order to generate a 32-bit output, the DES function applies a 48-bit key to the right-most 32 bits.



Details of Single Round of DES Algorithm:



1. The Primary Conversion

Permutation of sub-key collection from the initial key

2. Permutation of Extension (E-table)

The right half is increased from 32-bits to 48-bits

3. Substitution of S-box

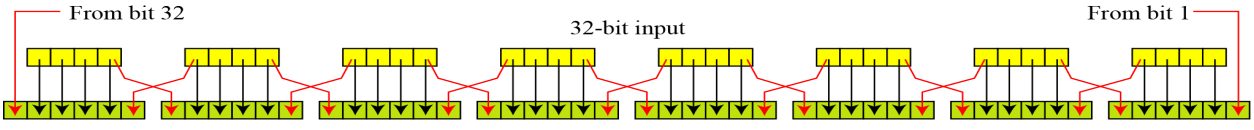
Accepts 48-bits from XOR operation and uses 8 substitution boxes to generate 32-bits (each S-box has a 6-bit i/p and 4-bit o/p).

4. Permutation by P-Box

5. XOR and Sharing

Expansion permutation:

Expansion P-box –E(R):Since RI−1 is a 32-bit input and KI is a 48-bit key, we first need to expand RI−1 to 48 bits.



32	01	02	03	04	05
04	05	06	07	08	09
08	09	10	11	12	13
12	13	14	15	16	17
16	17	18	19	20	21
20	21	22	23	24	25
24	25	26	27	28	29
28	29	31	31	32	01

Expansion P-box table: While it is mathematically possible to describe the relationship between the input and output, DES uses Table 2 to define this P-box.

Block Cipher Components:

P Box:

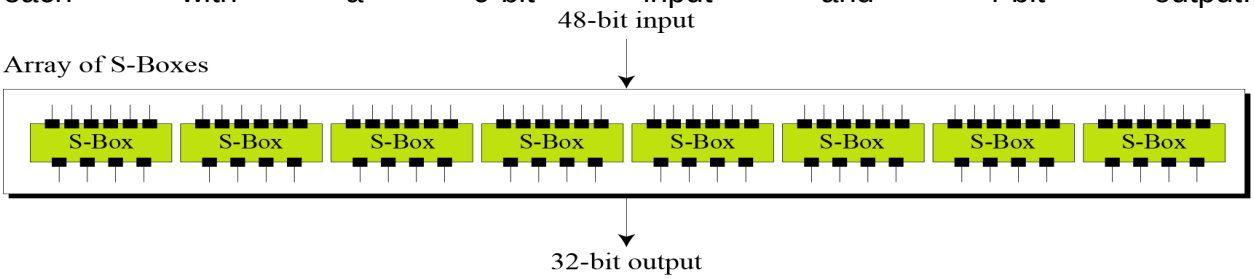
- Main cipher with less fixed transposition.
- Used to deliver diffusion
- Input bits are allowed to generate output bits
- Three forms are classified:

Whitener (XOR):

DES utilizes the XOR operation on the extended right portion and the round key after the expansion permutation. Note that 48-bits in length are both the correct section and the key. Note also that only in this process will the round key be used.

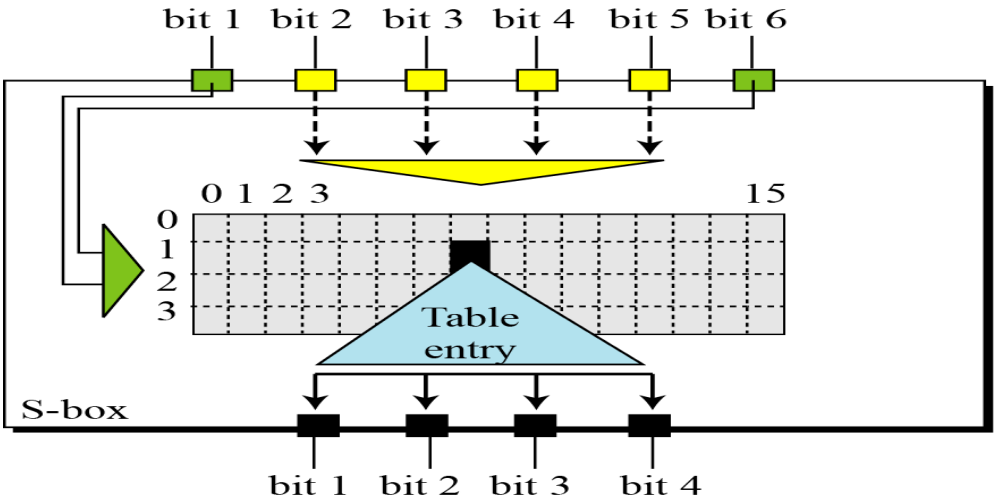
S-Boxes:

The actual blending is done by the S-boxes (confusion). Eight S-boxes are used by DES, each with a 6-bit input and 4-bit output.



	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	14	04	13	01	02	15	11	08	03	10	06	12	05	09	00	07
1	00	15	07	04	14	02	13	10	03	06	12	11	09	05	03	08
2	04	01	14	08	13	06	02	11	15	12	09	07	03	10	05	00
3	15	12	08	02	04	09	01	07	05	11	03	14	10	00	06	13

Table shows the permutation for S-box 1. For the rest of the boxes see the textbook.



Block Cipher Components:

S Box

- Key-less fixed cipher replacement.
- Used to provide uncertainty
- Based on unrecognized key
- Take n bit of plaintext as an input and generate m bit of cipher text as an output, where n & m may be the same or different value
- Mapping is set by default.

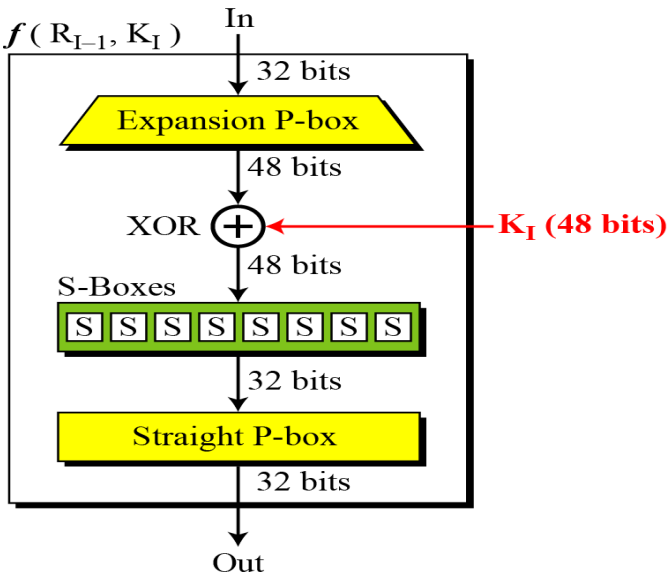
Example-1:

The S-box 1 input is 100011. Whose performance is it?

Solution: If we write together the first and sixth bits, we get 11 in binary, which is 3 in decimal, respectively. The bits that remain are 0001 in binary, which is 1 in decimal. In Table, we are searching for the value in row 3, column 1 (S-box 1). The consequence of this is 12 in decimal, which is 1100 in binary. So input 100011 yields 1100 output.

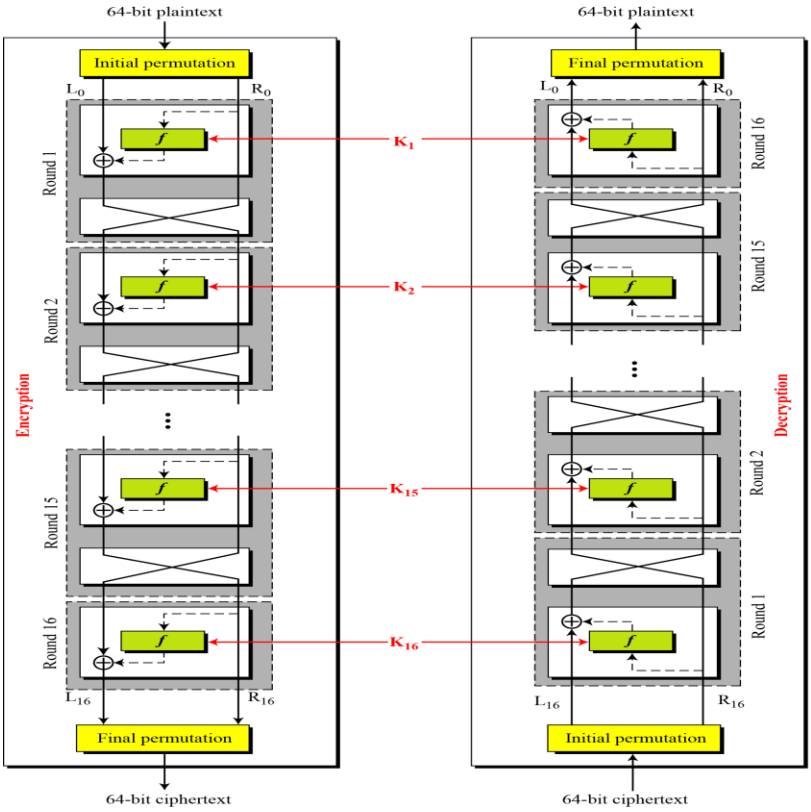
Straight Permutation:

16	07	20	21	29	12	28	17
01	15	23	26	05	18	31	10
02	08	24	14	32	27	03	09
19	13	30	06	22	11	04	25



- **Straight P box**
  - Takes N bits input permutes n bit output.
  - So n! Ways to map from i/p to o/p

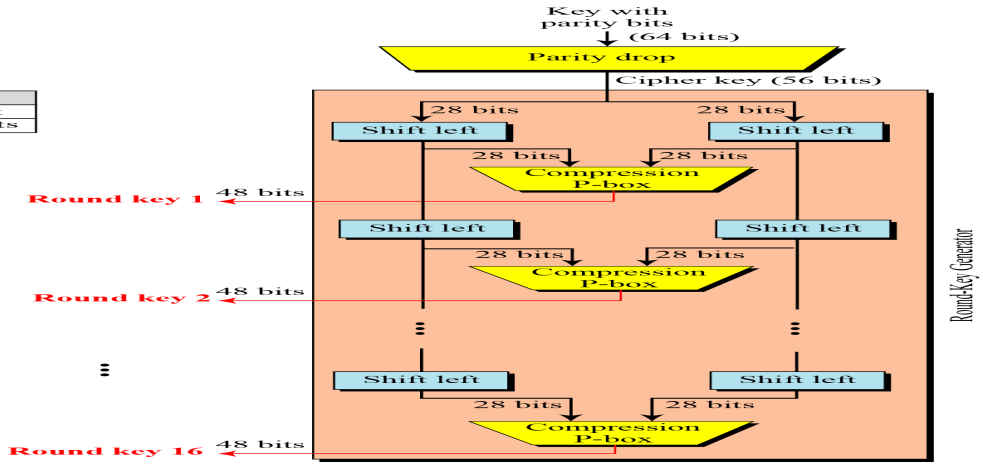
DES cipher and reverse cipher for the first approach:



Key Generation

Shifting

Rounds	Shift
1, 2, 9, 16	one bit
Others	two bits



The round-key generator creates sixteen 48-bit keys out of a 56-bit cipher key.

Parity-bit drop table PC-1:



57	49	41	33	25	17	09	01
58	50	42	34	26	18	10	02
59	51	43	35	27	19	11	03
60	52	44	36	63	55	47	39
31	23	15	07	62	54	46	38
30	22	14	06	61	53	45	37
29	21	13	05	28	20	12	04

Block Cipher Components:

Circular shift Classifies in two ways:

- Circular left shift: every bit of the word is pushed in the left direction by a particular number of positions
- Number of bits to the left are removed and put to the right Number of bits shifts

Round	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Bit shifts	1	1	2	2	2	2	2	2	1	2	2	2	2	2	2	1

Key-compression table (PC-2)

14	17	11	24	01	05	03	28
15	06	21	10	23	19	12	04
26	08	16	07	27	20	13	02
41	52	31	37	47	55	30	40
51	45	33	48	44	49	39	56
34	53	46	42	50	36	29	32

Example-3:



We choose a random plaintext block and a random key, and determine what the ciphertext block would be (all in hexadecimal):

Plaintext: 123456ABCD132536 Key: AABB09182736CCDD
CipherText: C0B7A8D05F3A829C

Plaintext: 123456ABCD132536			
*After initial permutation:*14A7D67818CA18AD After splitting: L_0=14A7D678 R_0=18CA18AD			
Round	*Left*	*Right*	*Round Key*
Round 1	18CA18AD	5A78E394	194CD072DE8C
Round 2	5A78E394	4A1210F6	4568581ABCCE
Round 3	4A1210F6	B8089591	06EDA4ACF5B5
Round 4	B8089591	236779C2	DA2D032B6EE3
Round 5	236779C2	A15A4B87	69A629FEC913
Round 6	A15A4B87	2E8F9C65	C1948E87475E
Round 7	2E8F9C65	A9FC20A3	708AD2DDB3C0
Round 8	A9FC20A3	308BEE97	34F822F0C66D
Round 9	308BEE97	10AF9D37	84BB4473DCCC
Round 10	10AF9D37	6CA6CB20	02765708B5BF
Round 11	6CA6CB20	FF3C485F	6D5560AF7CA5
Round 12	FF3C485F	22A5963B	C2C1E96A4BF3
Round 13	22A5963B	387CCDAA	99C31397C91F
Round 14	387CCDAA	BD2DD2AB	251B8BC717D0
Round 15	BD2DD2AB	CF26B472	3330C5D9A36D
Round 16	19BA9212	CF26B472	181C5D75C66D
After combination: 19BA9212CF26B472			
Ciphertext: C0B7A8D05F3A829C *(after final permutation)*			

DES ANALYSIS:

A strong magnifier has been used by critics to evaluate DES. In a block cipher, experiments have been carried out to measure the strength of certain desired properties.

Discussed topics in this section:



• Properties Properties

Criteria for Architecture

• DES Vulnerabilities

Properties: The avalanche effect and completeness are two desirable characteristics of a block cipher.

The Avalanche Effect : A beneficial aspect of any encryption algorithm is that a minor change in the ciphertext-avalanche effect can produce a substantial change in either the plaintext or the key.

In particular, in several bits of the ciphertext, a change in one bit of the plaintext or one bit of the key can cause a change.

Table 3.6 Avalanche Effect in DES: Change in Plaintext

Round		δ
	02468aceeca86420 12468aceeca86420	1
1	3cf03c0fbad22845 3cf03c0fbad32845	1
2	bad2284599e9b723 bad3284539a9b7a3	5
3	99e9b7230bae3b9e 39a9b7a3171cb8b3	18
4	0bae3b9e42415649 171cb8b3ccaca55e	34
5	4241564918b3fa41 ccaca55ed16c3653	37
6	18b3fa419616fe23 d16c3653cf402c68	33
7	9616fe2367117cf2 cf402c682b2cefbc	32
8	67117cf2c11bfc09 2b2cefbc99f91153	33

Round		δ
9	c11bfc09887fbc6c 99f911532eed7d94	32
10	887fbc6c600f7e8b 2eed7d94d0f23094	34
11	600f7e8bf596506e d0f23094455da9c4	37
12	f596506e738538b8 455da9c47f6e3cf3	31
13	738538b8c6a62c4e 7f6e3cf34bc1a8d9	29
14	c6a62c4e56b0bd75 4bc1a8d91e07d409	33
15	56b0bd7575e8fd8f 1e07d4091ce2e6dc	31
16	75e8fd8f25896490 1ce2e6dc365e5f59	32
IP^{-1}	da02ce3a89ecac3b 057cde97d7683f2a	32



Table 3.7 Avalanche Effect in DES: Change in Key

Round		δ
	02468aceeca86420 02468aceeca86420	0
1	3cf03c0fbad22845 3cf03c0f9ad628c5	3
2	bad2284599e9b723 9ad628c59939136b	11
3	99e9b7230bae3b9e 9939136b768067b7	25
4	0bae3b9e42415649 768067b75a8807c5	29
5	4241564918b3fa41 5a8807c5488dbe94	26
6	18b3fa419616fe23 488dbe94aba7fe53	26
7	9616fe2367117cf2 aba7fe53177d21e4	27
8	67117cf2c11bfc09 177d21e4548f1de4	32

Round		δ
9	c11bfc09887fbc6c 548f1de471f64dfd	34
10	887fbc6c600f7e8b 71f64dfd4279876c	36
11	600f7e8bf596506e 4279876c399fdc0d	32
12	f596506e738538b8 399fdc0d6d208dbb	28
13	738538b8c6a62c4e 6d208dbbb9bdeeaa	33
14	c6a62c4e56b0bd75 b9bdeeaad2c3a56f	30
15	56b0bd7575e8fd8f d2c3a56f2765c1fb	33
16	75e8fd8f25896490 2765c1fb01263dc4	30
IP^{-1}	da02ce3a89ecac3b ee92b50606b62b0b	30

Completeness effect:

The effect of completeness means that each bit of the ciphertext needs to depend on the plaintext for several bits.

Criteria of Design:

S-Boxe:

From one round to the next, the design offers uncertainty and diffusion of parts.

P-The boxes:

They give the distribution of bits.

Number of Rounds:

Feistel ciphers are used by DES for sixteen rounds. The ciphertext is a thoroughly random feature of plaintext and ciphertext.

DES Weaknesses:

Some flaws in DES have been noticed by critics over the last few years.

Cipher Interface vulnerabilities



1. In S-boxes, vulnerabilities

2. P-boxes vulnerabilities

3. Deficiencies in the Main

Security of DES:

DES has gone through much scrutiny as the first important block cipher. Three are of note among the attacks attempted: brute-force, differential cryptanalysis, and linear cryptanalysis.

Discussed topics in this section:

Assault by Brute-Force

Cryptanalysis Differential

Linear Cryptanalysis •

The vulnerability of the short cipher key in DES has been solved. Combining this weakness with the key complement weakness, it is obvious that using 255 encryptions, DES can be broken.

Differential Cryptanalysis: It was found that DES designers were already aware of this type of attack and constructed S-boxes and selected 16 as the number of rounds to make DES strictly resistant to this type of attack.

Linear Cryptanalysis:

Linear cryptanalysis is newer than cryptanalysis of differentials. DES is more vulnerable to linear cryptanalysis than to cryptanalysis of variations. With linear cryptanalysis, S-boxes are not very resistant. It has been shown that, using 243 pairs of known plaintexts, DES can be broken. Finding so many sets, however is very unlikely from a realistic point of view.

The Strength of DES:

- The use of 56-bit keys

There are 256 possible keys with a key length of 56 bits, which is roughly 7.2x1016 keys. Thus, brute-force attack seems unrealistic on the face of it.



In July 1998, when the Electronic Frontier Foundation (EFF) revealed that it had broken DES encryption using a special-purpose "DES Crackers" computer that was designed for less than $250,000, DES finally and certainly proved unsafe.

- There are, luckily, a variety of alternatives to DES, of which AES and triple DES are the most common.

The nature of the DES Algorithm.

The possibility that cryptanalysis is possible by leveraging the characteristics of the DES algorithm is another concern.

The eight substitution tables, or S-boxes, that are used in each iteration have been the object of concern.

Since the design requirements were not made available for these boxes, and indeed for the entire algorithm, there is a mistrust that the boxes were designed in such a way that cryptanalysis is feasible for an opponent who knows the weakness of the S-boxes.

Even if this, no one has so far succeeded in discovering the supposed fatal weaknesses in the S-boxes.

- **Timing Attacks**

Timing attack is one in which the key or plaintext information is obtained by calculating how long it takes for a specific implementation to decrypt different ciphertext.

The fact that an encryption or decryption algorithm sometimes takes slightly different amounts of time on different inputs is exploited by a timing attack.

A good timing attack tends to be reasonably immune to DES.

Block cipher design principles:

- DES design criteria

Emphasis on S box and P role design

Number of rounds



The greater the number of rounds, even for a relatively weak F, the more difficult it is to perform cryptanalysis.

Development of the F-function

- The algorithm should have strong avalanche properties.
- Bit independence criterion (BIC) which states that when any single bit I is inverted, the output bits j and k should change independently.
- Main algorithm schedule.

Select the subkeys to maximize the difficulty of deducing individual subkeys and the difficulty of working back to the main key.

Differential and Linear Cryptanalysis:

- Differential attack Cryptanalysis
- It was discovered in 1990.
- 247 selected plain texts were required.
- The first published attack that is capable of cracking DES with less than 255 encryptions is differential cryptanalysis.
- The concept behind differential cryptanalysis is to observe, instead of observing the evolution of a single text block, the action of pairs of text blocks evolving along each round of the cipher.

Differential attack Cryptanalysis

Consider the initial plaintext block m to be two halves of m0.m1.

Each round of DES maps the right-hand input into the left-hand output and sets the right-hand output to be a function of the left-hand input and the subkey for this round. So, in each round, only one new 32-bit block is created.

If each new block mi (2 <= i<= 17) is labelled, then the intermediate message halves are linked as follows:

$$m_{i+1} = m_{i-1} \oplus \mathrm{f}(m_i, K_i), \qquad i = 1, 2, \ldots, 16$$

Differential Cryptanalysis attack:

In differential cryptanalysis, we start two messages, m and m', with a known XOR difference ∆m=m XOR m' and consider the difference between the intermediate message halves: ∆mi=mi XOR mi' then we have



$$\begin{aligned}\Delta m_{i+1} &= m_{i+1} \oplus m'_{i+1} \\ &= [m_{i-1} \oplus \mathrm{f}(m_i, K_i)] \oplus [m'_{i-1} \oplus \mathrm{f}(m'_i, K_i)] \\ &= \Delta m_{i-1} \oplus [\mathrm{f}(m_i, K_i) \oplus \mathrm{f}(m'_i, K_i)]\end{aligned}$$

So if we do, and we know it with a high probability, then we know it with a high probability. In addition, when a number of such variations are calculated, it is feasible to evaluate the subkey used in function F.

Linear Cryptanalysis:

- It is a slight improvement over cryptanalysis differentials.
- This attack is based on the discovery of linear approximations to describe the transformations carried out in DES.
- Compared to 247 selected plaintexts for differential cryptanalysis, this method can discover a DES key given 243 known plaintexts.
- While this is a slight change, as an assault on DES, it still makes linear cryptanalysis unworkable.

Linear Cryptanalysis:

We now give a brief summary of the principle on which linear cryptanalysis is based. For a cipher with n-bit plaintext and ciphertext blocks and an m-bit key, let the plaintext block be labeled $\mathrm{P}[1], \ldots \mathrm{P}[n]$, the cipher text block $\mathrm{C}[1], \ldots \mathrm{C}[n]$, and the key $\mathrm{K}[1], \ldots, \mathrm{K}[m]$. Then define

$$\mathrm{A}[i, j, \ldots, k] = \mathrm{A}[i] \oplus \mathrm{A}[j] \oplus \ldots \oplus \mathrm{A}[k]$$

The objective of linear cryptanalysis is to find an effective *linear* equation of the form:

$$\mathrm{P}[\alpha_1, \alpha_2, \ldots, \alpha_a] \oplus \mathrm{C}[\beta_1, \beta_2, \ldots, \beta_b] = \mathrm{K}[\gamma_1, \gamma_2, \ldots, \gamma_c]$$

Once a proposed relation is determined, the procedure is to compute the results of the left-hand side of the preceding equation for a large number of plain text–ciphertext pairs. If the result is 0 more than half the time, assume $\mathrm{K}[\gamma_1, \gamma_2, \ldots, \gamma_c] = 0$. If it is 1 most of the time, assume $\mathrm{K}[\gamma_1, \gamma_2, \ldots, \gamma_c] = 1$.

Block Cipher Modes of Operations:

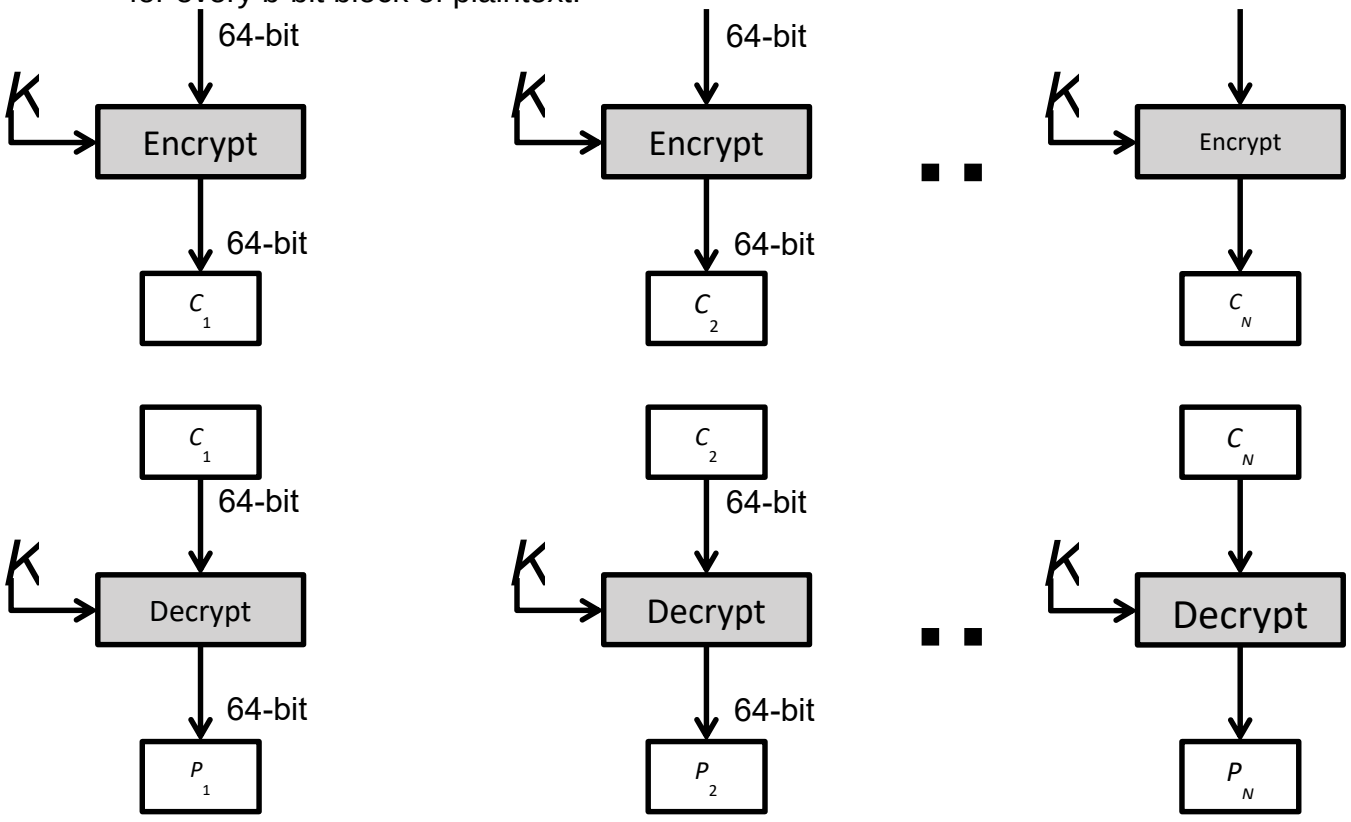
- Five "modes of operation" were specified to apply a block cipher in a variety of applications.
- The five modes are intended to cover a wide range of encryption applications for which a block cipher could be used.

- These modes are intended for use, like triple DES and AES, with any symmetric block cipher.

Electronic Code Book (ECB)
Cipher Block Chaining (CBC)
Cipher Feedback (CFB)
Output Feedback (OFB)
Counter (CTR)

1. Electronic Code Book (ECB)

- In ECB Mode Plaintext handled one block at a time and each block of plaintext is encrypted using the same key.
- The term codebook is used because, for a given key, there is a unique ciphertext for every b-bit block of plaintext.



- Strength: it's simple.
- Weakness:

Repetitive information contained in the plaintext may show in the ciphertext, if aligned with blocks.

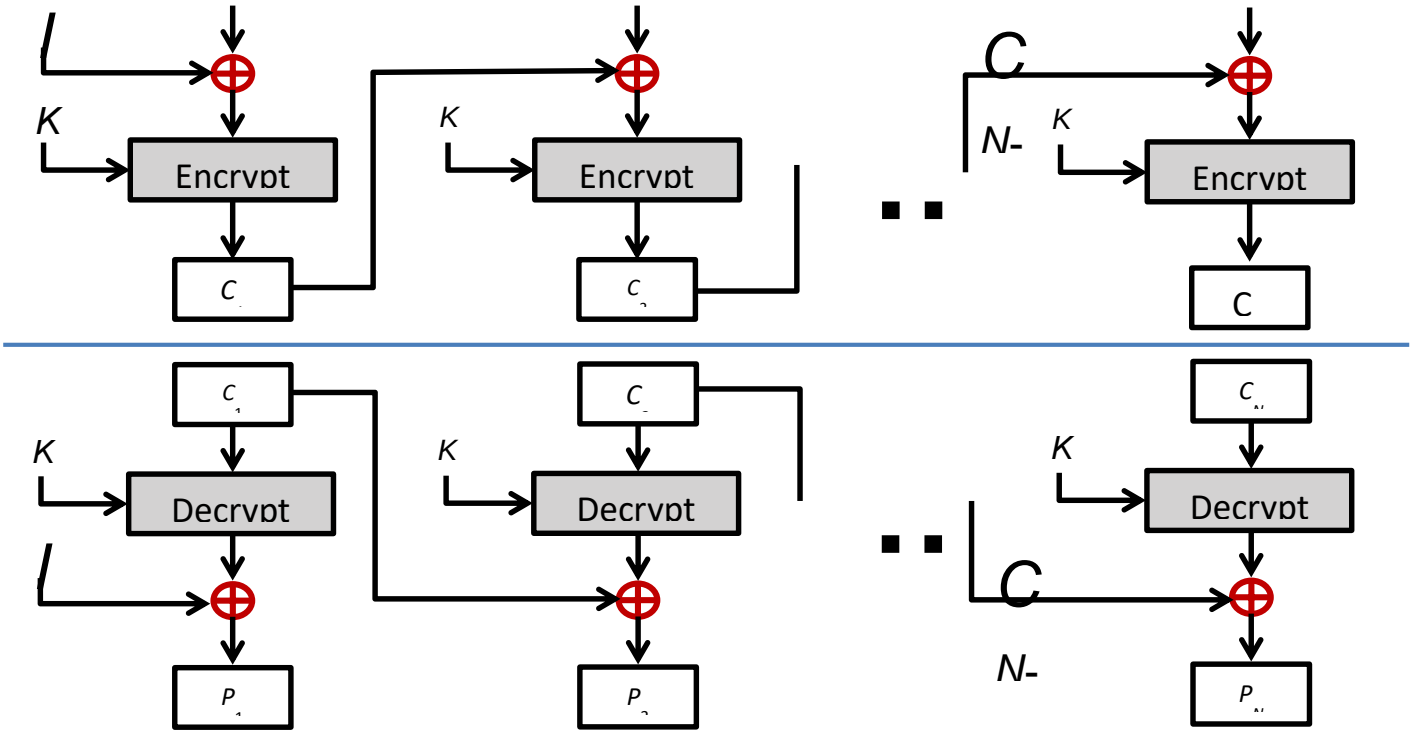
If the message has repetitive elements with a period of repetition a multiple of b bits, then these elements can be identified by the analyst.

- Typical application:

Secure transmission of short pieces of information (e.g. a temporary encryption key)

2. Cipher Block Chaining (CBC):

- CBC is a technique in which different ciphertext blocks are generated by the same plaintext block, if repeated.
- In this scheme, the XOR of the current plaintext block and the preceding ciphertext block are the input to the encryption algorithm; the same key is used for each block.
- An initialization vector (IV) is XORed with the first block of plaintext to generate the first block of ciphertext.
- On decryption, the IV is XORed with the output of the decryption algorithm to recover the first block of plaintext.

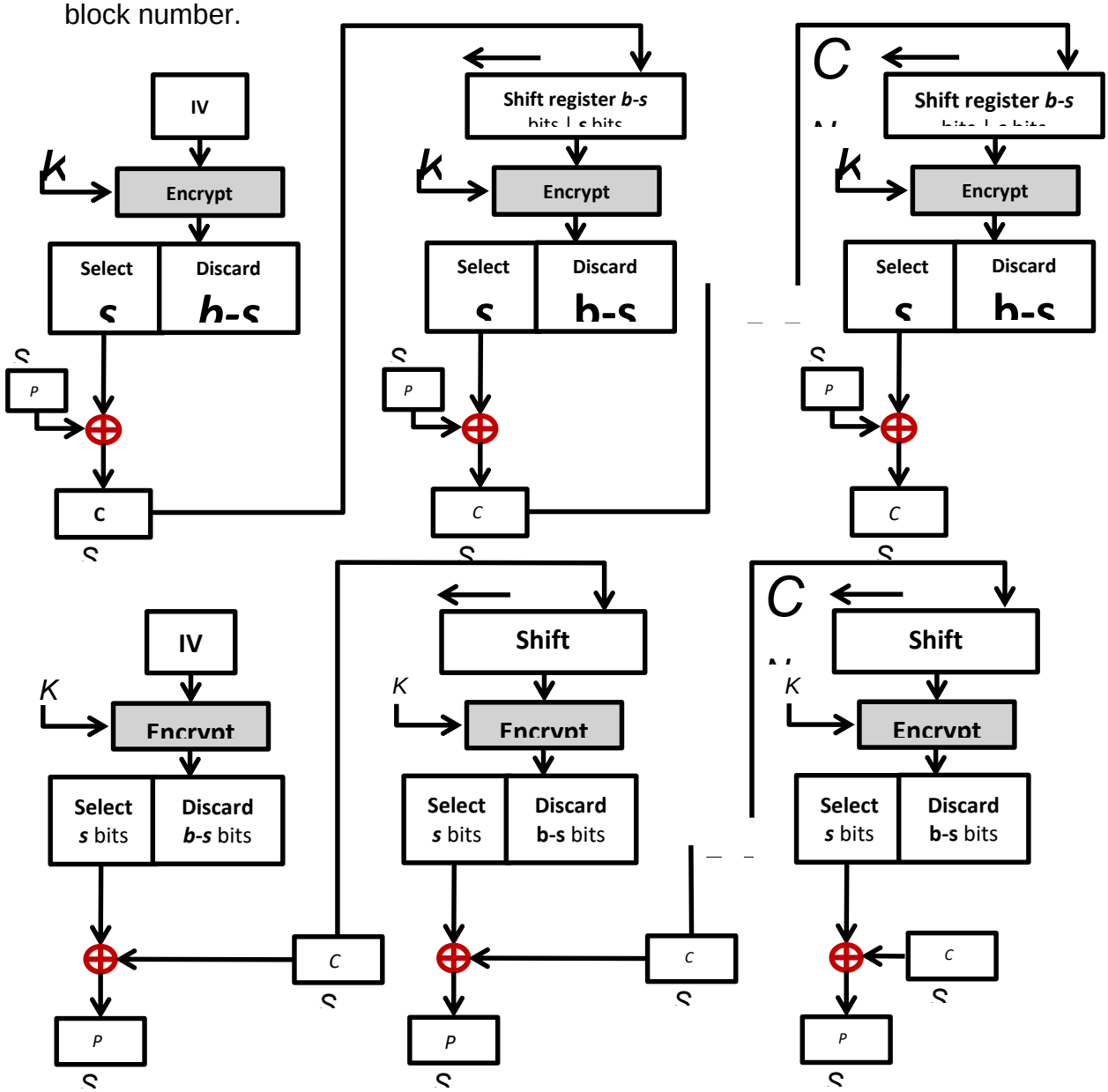


Strength: because of the chaining mechanism of CBC, it is an appropriate mode for encrypting messages of length greater than b bits

- Typical application: General-purpose block-oriented transmission Authentication

3. Cipher Feedback Mode (CFB)

- Encryption is achieved on a block of bits for AES, DES, or any block cipher. B = 64 in DES, and b = 128 in AES.
- However, using cipher feedback (CFB) mode, output feedback (OFB) mode, and counter (CTR) mode, it is possible to transform a block cipher into a stream cipher.
- A stream cipher removes the need to pad a message in order to be an integral block number.



CFB Mode:

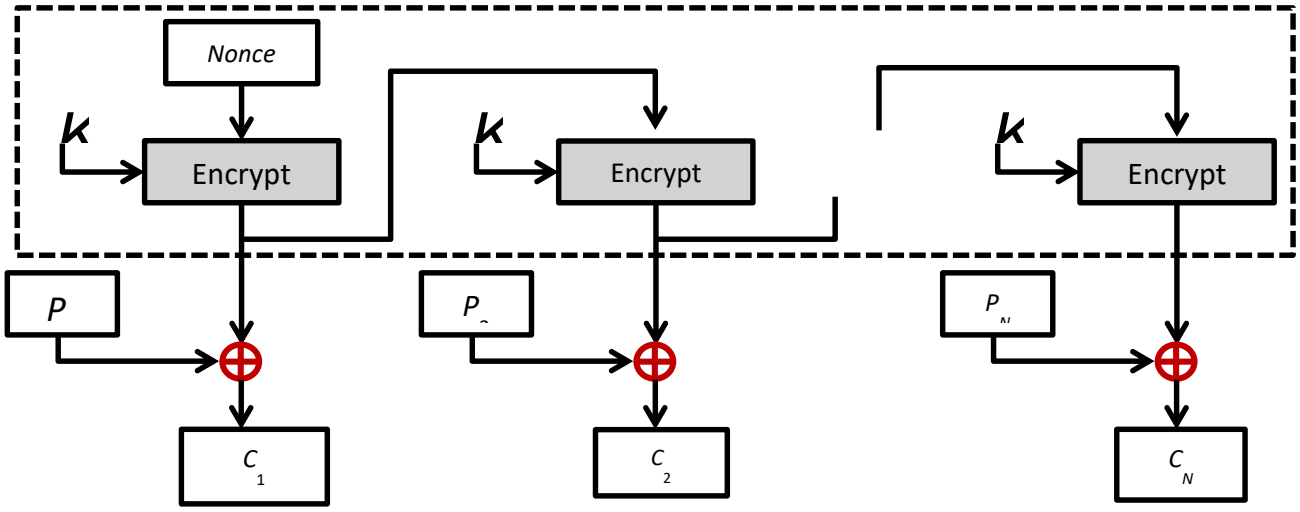


- A b-bit shift register that is initially set to some initialization vector is the input to the encryption function (IV).
- XORed with the first segment of plaintext P1 to generate the first unit of ciphertext C1, which is then transmitted, is the left (most important bits of the output of the encryption function.
- Furthermore, the contents of the shift register are shifted by s bits to the left and C1 is put in the shift register's rightmost (least significant) s bits.
- For decryption, the same scheme is used, except that the received ciphertext unit is XORed with the output of the encryption function to produce the plaintext unit.

CFB	$I_1 = IV$ $I_j = \text{LSB}_{b-s}(I_{j-1}) \| C_{j-1}$ $j = 2, \ldots, N$ $O_j = \text{E}(K, I_j)$ $j = 1, \ldots, N$ $C_j = P_j \oplus \text{MSB}_s(O_j)$ $j = 1, \ldots, N$	$I_1 = IV$ $I_j = \text{LSB}_{b-s}(I_{j-1}) \| C_{j-1}$ $j = 2, \ldots, N$ $O_j = \text{E}(K, I_j)$ $j = 1, \ldots, N$ $P_j = C_j \oplus \text{MSB}_s(O_j)$ $j = 1, \ldots, N$

4. Output Feedback Mode (OFB)

- The output feedback (OFB) mode is structurally similar to the CFB mode.
- For OFB, the encryption function output is fed back to become the input for the next block of plaintext to be encrypted.
- In CFB, the XOR unit output is fed back to become an entry for the next block to be encrypted.
- The other distinction is that the OFB mode operates on complete plaintext and ciphertext blocks, while the CFB operates on a subset of s-bits.
- Nonce: A time-varying value that has at most a negligible chance of repeating, for example, a random value that is generated anew for each use, a timestamp, a sequence number or some combination of these.



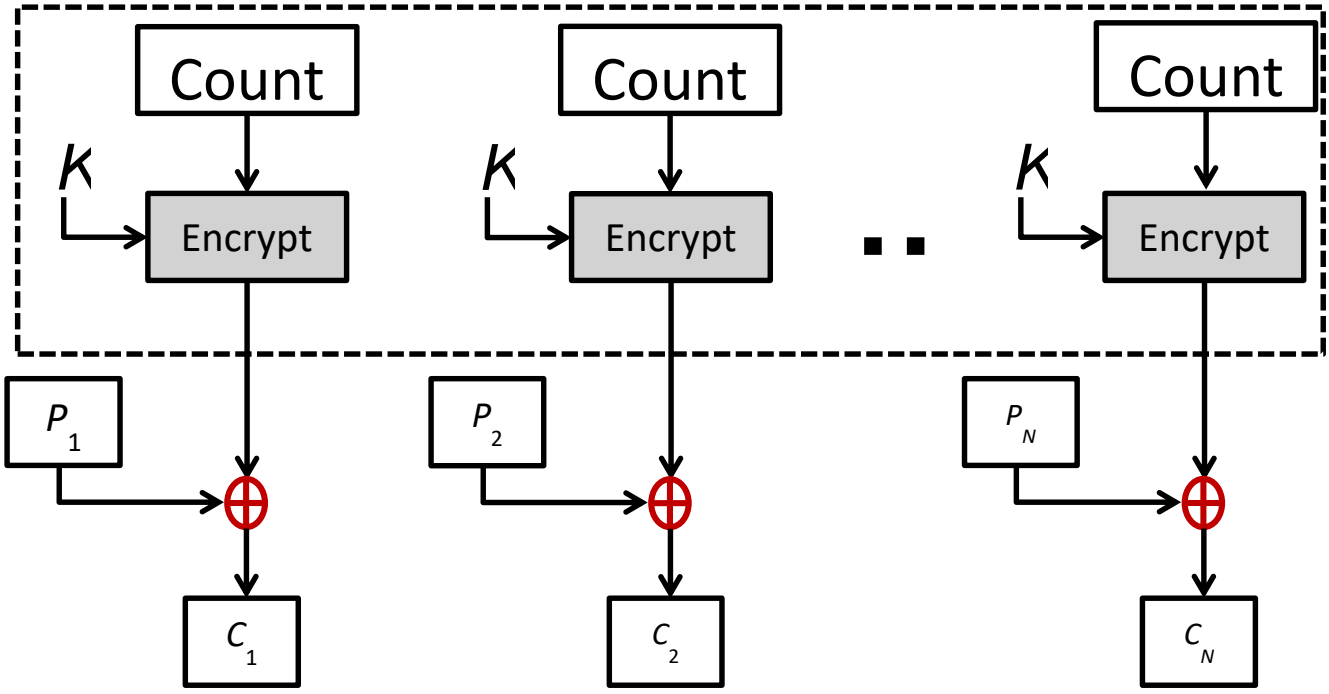
OFB Mode:

- Each bit in the ciphertext is independent of the previous bit or bits.
- This avoids error propagation
- Pre-compute of forward cipher is possible

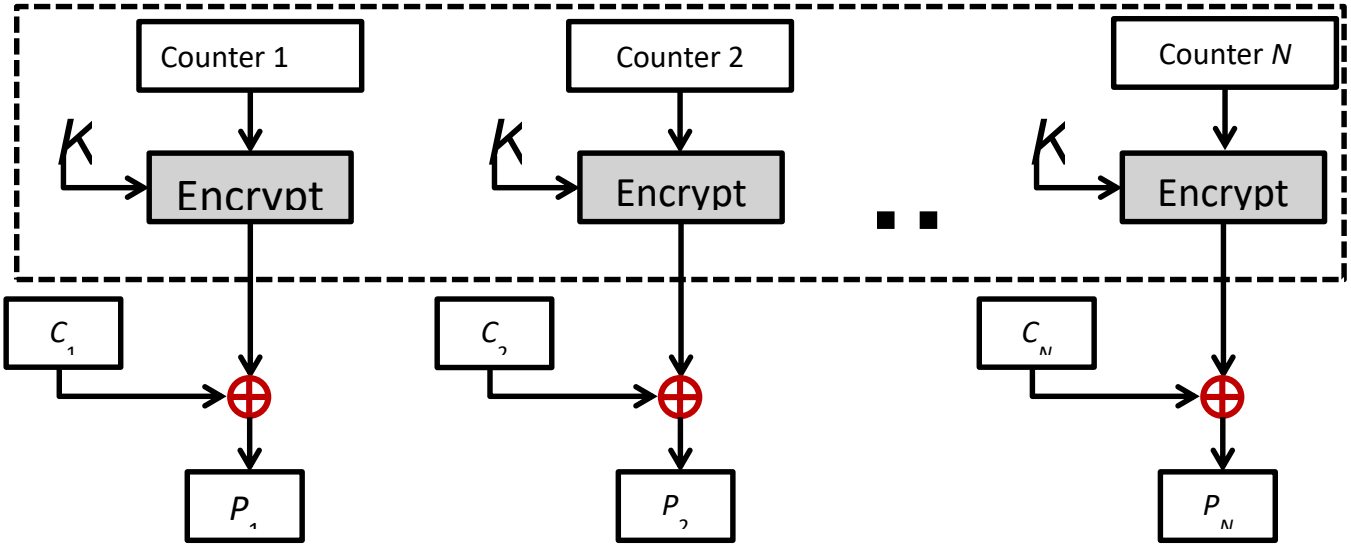
OFB	$I_1 = Nonce$ $I_j = O_{j-1} \quad j = 2, \ldots, N$ $O_j = \mathrm{E}(K, I_j) \quad j = 1, \ldots, N$ $C_j = P_j \oplus O_j \quad j = 1, \ldots, N-1$ $C_N^* = P_N^* \oplus \mathrm{MSB}_u(O_N)$	$I_1 = Nonce$ $I_j = O_{j-1} \quad j = 2, \ldots, N$ $O_j = \mathrm{E}(K, I_j) \quad j = 1, \ldots, N$ $P_j = C_j \oplus O_j \quad j = 1, \ldots, N-1$ $P_N^* = C_N^* \oplus \mathrm{MSB}_u(O_N)$

5. Counter Mode (CTR):

- Counter (CTR) mode has recently increased with network protection and IP sec applications for ATM (asynchronous transfer mode) (IP security).
- A counter equal to the size of the plaintext block is used.
- For each plaintext block that is encrypted, the counter value has to be different.
- The counter is normally initialized to a certain value and then increased by 1 for each subsequent block.



CTR Decryption:



CTR	$C_j = P_j \oplus \mathrm{E}(K, T_j) \quad j = 1, \ldots, N-1$ $C_N^* = P_N^* \oplus \mathrm{MSB}_u[\mathrm{E}(K, T_N)]$	$P_j = C_j \oplus \mathrm{E}(K, T_j) \quad j = 1, \ldots, N-1$ $P_N^* = C_N^* \oplus \mathrm{MSB}_u[\mathrm{E}(K, T_N)]$

Advantages of the CTR Mode:

• Strengths:

The encryption algorithm only requires

Random access to data blocks that are encrypted

Blocks may be processed in parallel (encrypted or decrypted).

Easy; rapid encryption/decryption

- Counters have to be

Unknown and unforeseeable must be



In the main stream, pseudo-randomness is a goal

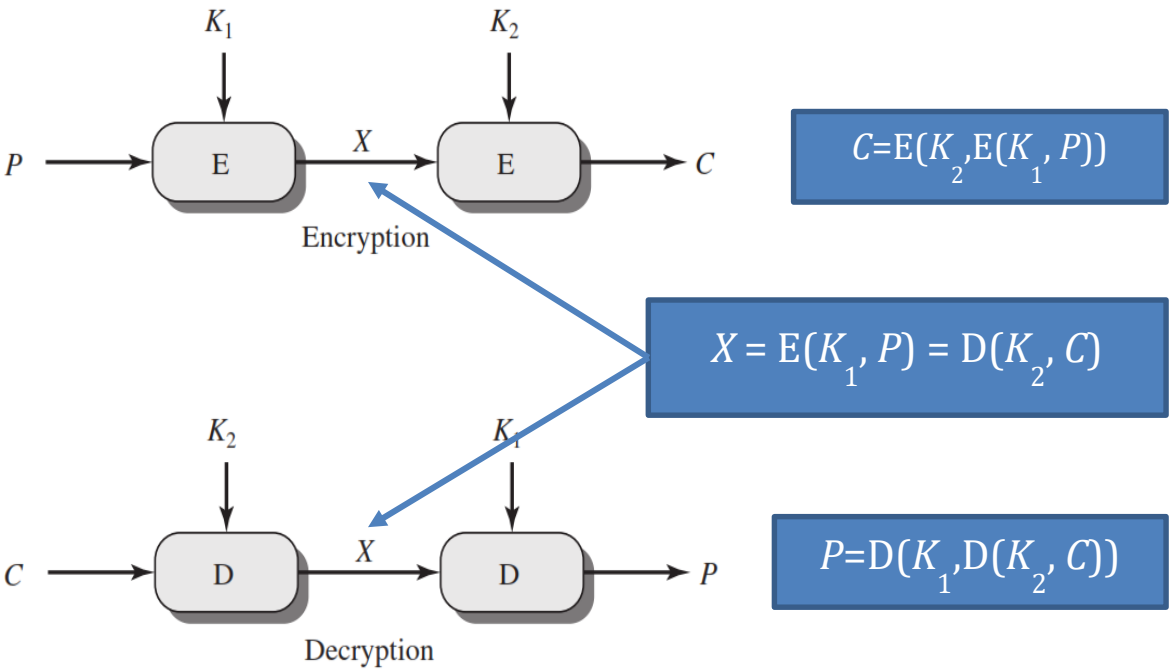
Summary of all modes

Operation Mode	Description	Type of Result
ECB	Each n-bit block is encrypted independently with same key	Block Cipher
CBC	Same as ECB, but each block is XORed with previous cipher text	Block Cipher
CFB	Each s-bit block is XORed with s-bit key which is part of previous cipher text	Stream Cipher
OFB	Same as CFB, but the shift register is updated by the previous s-bit key	Stream Cipher
CTR	Same as OFB, but a counter is used instead of nonce	Stream Cipher

Multiple Encryption:

- There has been significant interest in seeking an alternative, given the possible weakness of DES to a brute-force attack.
- One solution is to develop a completely new algorithm, a prime example of which is AES.
- The use of multiple encryption with DES and multiple keys is another option that will sustain the current investment in software and equipment.

Double DES:

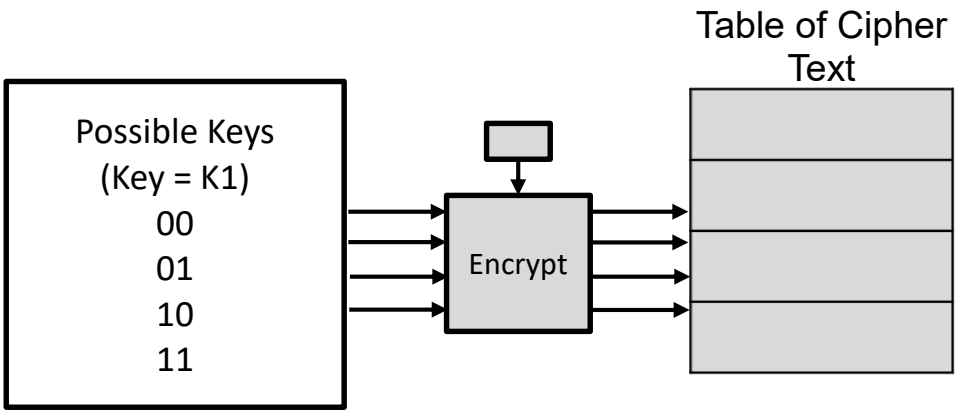


Meet in the Middle Attack:

- This attack involves encryption from one end, decryption from the other and matching the results in the middle.
- Suppose cryptanalyst knows P and corresponding C.
- Now, the aim is to obtain the values of K1 and K2.

Meet in the Middle Attack Step-1

- For all possible values (256) of key K1, the cryptanalyst would encrypt the P by performing E(K1,P).
- The cryptanalyst would store output in a table.





What is RC4 Encryption?

- RC4 means Rivest Cipher 4, which Ron Rivest invented for RSA Security in 1987. These are Stream Ciphers. Stream Ciphers function on a byte by byte stream of data. Thanks to its simplicity and speed of operation, the RC4 stream cipher is one of the most commonly used stream ciphers. It is a stream cipher of variable key-size with byte-oriented operations. Either 64 bit or 128-bit key sizes are used. In applications such as Stable Socket Layer (SSL), Transport Layer Security (TSL),Transport Layer Security (TSL) and also used in IEEE 802.11 wireless LAN std. it is commonly used.
- Why Encryption Is Important?

  It is possible to avoid unauthorized access to data through encryption. Third parties do not have access to data that we exchange or receive while we execute encryption. The encryption is achieved using a hidden key, or we can use a public key and a private key to say that. Both the sender and the recipient have their public key and private key from which plain text encryption and ciphertext decryption is carried out.

  Types of RC4

  - There are different RC4 forms, such as Spritz, RC4A, VMPC, and RC4A.
  - SPRITZ: Spritz can be used to construct a cryptographic hash function, a Deteristic Random Bit Generator (DRBG), and an encryption algorithm that supports authenticated data encryption (AEAD).
  - RC4A: An RC4 version has been suggested by Souraduyti Paul and Bart Preneel, which they call RC4A, which is stronger than RC4.
  - VMPC: VMPC is another RC4 version that stands for Permutation Composition Variably Modified.
  - RC4A+: RC4A+ is a modified version of RC4 with a more complex three-phase key schedule that takes approximately three times as long as RC4 and a more complex output feature that performs four additional lookups for each byte output in the S array, taking approximately 1.7 times as long as the basic RC4.
  - Algorithm: The algorithm works on 1 to 256 bytes (8 to 2048 bits) of a user-selected variable-length key(K), usually between 5 and 16 bytes. The master key is used for generating a 256-byte state vector S.The first step is the initialisation of the collection. It is a 256-size character string, i.e. About S[256]. After that, we initialize S[i] to I for every element of the array.



Code for array initialization:

```
Char S[256];

int i;

for(i=0;i<256;i++)

S[i] = i
```

The array will look like -

S[] = {0, 1, 2, 3, ------, 254, 255}

After this, we will run the **KSA algorithm-**

KSA is going to use the secret key to scramble this array. KSA is a simple loop, in which we are having two variable i and j. We are using these variables to rearrange the array. Rearranging the array is done by using a secret key.

Code for KSA (Key Scheduling Algorithm) :

```
int i, j=0;

for(i=0;i<256;i++)

{

j=( j + S[i] + T[i]) mod 256;

Swap(S[i], S[j]);

}
```

KSA has been scrambled, S[256] array is used to generate the PRGA(Pseudo Random Generation Algorithm). This is the actual Keystream.

Code for PRGA (Pseudo Random Generation Algorithm):

```
i=j=0;

while(true)
```



{

i = (i + 1) mod 256;

j = (j + S[i]) mod 256;

Swap(S[i], S[j]);

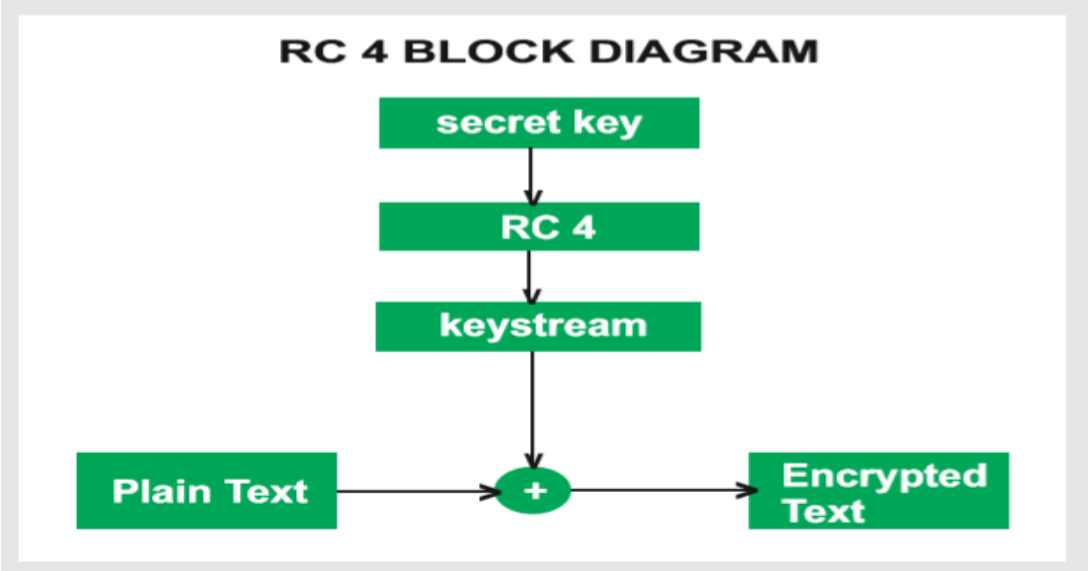
t = (S[i] + S[j]) mod 256 ;

k = S[t];

}

This is the next step of scrambling.

RC4 Block Diagram



Working of RC4

- • Method for encryption
- A plain text file and a hidden key are entered by the user.
- By using KSA and PRGA Algorithm, the encryption engine then generates the keystream.
- This keystream is now XOR with the plain text, to generate the encrypted text, this XORing is performed byte-by-byte.



- The encrypted text is then sent to the intended recipient, then the text is decrypted by the intended recipient and the original plain text is received by the recipient after decryption.
- Decryption Procedure
- Decryption is achieved by doing the same byte-wise X-OR operation on the Ciphertext.

Example: Let A be the plain text and B be the keystream (A xor B) xor B = A

Advantages

- RC4 stream ciphers are easy to use.
- In comparison with other ciphers, the speed of operation in RC4 is high.
- In coding, RC4 stream ciphers are efficient and simple to implement.
- RC4 stream ciphers need no more memory.
- On wide streams of data, RC4 stream ciphers are introduced.

Disadvantages

- If RC4 with a powerful MAC is not used, then encryption is vulnerable to a bit-flipping attack.

- Authentication is not supported by RC4 stream ciphers.

- Before new systems are used the RC4 algorithm needs additional research.
- RC4 stream ciphers on small data streams can not be introduced.
- RC4 does not discard the start of the output keystream or fails to use the algorithm's non-random or linked keys.

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